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COURSE : Data Structure

COURSE CODE : CSA0311

CO2 - AT1 Analytical problem
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Ans.
2/19.

Consider the Singly linked list.

$10 \rightarrow 10 \rightarrow 30 \rightarrow 40 \rightarrow 50$

Delete the node containing 30. Explain how the links change deletion.

○ Delete the node containing 30

Initial linked list:

$10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50$

Steps:

1. Find the node containing 30.
2. The previous node is 20.
3. Change the link of 20 from 30 to 40.
4. Delete the node 30.

After Deletion :

10 $\rightarrow$ 20 $\rightarrow$ 40 $\rightarrow$ 50

Link Change :

○ Before : 20 $\rightarrow$ 30 $\rightarrow$ 40

○ After : 20 $\rightarrow$ 40

2) Merge the two Sorted Linked List

Given :

L1 : 1 $\rightarrow$ 3 $\rightarrow$ 5 $\rightarrow$ 7

L2 : 2 $\rightarrow$ 4 $\rightarrow$ 6 $\rightarrow$ 8

Merged Sorted Linked List :

1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$ 5 $\rightarrow$ 6 $\rightarrow$ 7 $\rightarrow$ 8